UDP CLIENT

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/socket.h>
#include <netinet/in.h>
#include <arpa/inet.h>
int main()
{
   int client;
   struct sockaddr_in servAddr;
   char servMsg[2000], cliMsg[2000];
   socklen_t server_struct_length = sizeof(servAddr);
   client = socket(AF_INET, SOCK_DGRAM, IPPROTO_UDP);
   if (client < 0)
   {
      printf("Error while creating socket\n");
      exit(1);
   }
   printf("Socket created successfully\n");
   memset(&servAddr, 0, sizeof(servAddr));
   servAddr.sin_family = AF_INET;
   servAddr.sin_port = htons(2002);
   servAddr.sin_addr.s_addr = inet_addr("127.0.0.1");
   printf("Enter message to server: ");
   fgets(cliMsg, sizeof(cliMsg), stdin);
   cliMsg[strcspn(cliMsg, "\n")] = '\0';
```

```
    if (sendto(client, cliMsg, strlen(cliMsg), 0,
            (struct sockaddr *)&servAddr,
            server_struct_length) < 0)
    {
        printf("Unable to send message\n");
        close(client);
        exit(1);
    }
    if (recvfrom(client, servMsg, sizeof(servMsg) - 1, 0,
             (struct sockaddr *)&servAddr,
             &server_struct_length) < 0)
    {
        printf("Error while receiving server's message\n");
        close(client);
        exit(1);
    }
    servMsg[sizeof(servMsg) - 1] = '\0';
    printf("Message from server: %s\n", servMsg);
    close(client);
    return 0;
}
```

UDP SERVER

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/socket.h>
#include <netinet/in.h>
#include <arpa/inet.h>
int main()
{
   int server;
   struct sockaddr_in servAddr, client_addr;
   char servMsg[2000], cliMsg[2000];
   socklen_t client_struct_length = sizeof(client_addr);
   server = socket(AF_INET, SOCK_DGRAM, IPPROTO_UDP);
   if (server < 0)
   {
      printf("Error while creating socket\n");
      exit(1);
   }
   printf("Socket created successfully\n");
   memset(&servAddr, 0, sizeof(servAddr));
   servAddr.sin_family = AF_INET;
   servAddr.sin_port = htons(2002);
   servAddr.sin_addr.s_addr = inet_addr("127.0.0.1");
   if (bind(server, (struct sockaddr *)&servAddr,
         sizeof(servAddr)) < 0)
   {
```

```
        printf("Couldn't bind to the port\n");
        close(server);
        exit(1);
    }
    printf("Binding done\n");
    printf("Listening done...\n");
    if (recvfrom(server, cliMsg, sizeof(cliMsg) - 1, 0,
                (struct sockaddr *)&client_addr,
                &client_struct_length) < 0)
    {
        printf("Couldn't receive\n");
        close(server);
        exit(1);
    }
    cliMsg[sizeof(cliMsg) - 1] = '\0';
    printf("Message from client: %s\n", cliMsg);
    strcpy(servMsg, cliMsg);
    if (sendto(server, servMsg, strlen(servMsg), 0,
              (struct sockaddr *)&client_addr,
              client_struct_length) < 0)
    {
        printf("Can't send\n");
        close(server);
        exit(1);
    }
    close(server);
    return 0;
}
```

OUTPUT

Client

Socket created successfully

Enter message to server: Hi this is client

Message from server: Hi this is client

Server

Socket created successfully

Binding done

Listening done...

Message from client: Hi this is client